

- 1. Install Nessus Essentials.**
- 2. Complete initial setup.**
- 3. Create a Basic Network Scan.**
- 4. Scan:**
 - **Localhost**
 - **Personal Test Machine**
 - **Lab Environment**

Observe:

- **Vulnerabilities Found**
- **Severity Levels**
- **CVSS Scores**
- **Affected Services**
- **Recommendations**

Assignment:

Create a report containing:

- 1. Critical Findings**
- 2. High Findings**
- 3. Medium Findings**
- 4. Low Findings**
- 5. Remediation Steps**
- 6. Screenshots**

Research the top 5 vulnerabilities detected and explain:

- **Impact**
- **Risk**
- **Fix**

1. Introduction

Nessus Essentials is a vulnerability assessment tool developed by Tenable. It is used to identify security weaknesses, misconfigurations, open ports, running services, and potential vulnerabilities present in a system or network. Nessus helps security professionals and administrators improve the security posture of their systems by providing detailed information about detected issues and remediation recommendations.

The objective of this assessment was to install Nessus Essentials, perform vulnerability scans on different targets, analyze the findings, and document the results.

2. Objectives

The objectives of this assignment were:

- Install Nessus Essentials.
- Complete the initial configuration.
- Create a Basic Network Scan.
- Scan localhost.
- Scan a personal test machine.
- Attempt scanning a lab environment.
- Analyze vulnerabilities found.
- Review severity levels and CVSS information.
- Identify affected services.
- Provide remediation recommendations.

3. Nessus Installation and Setup

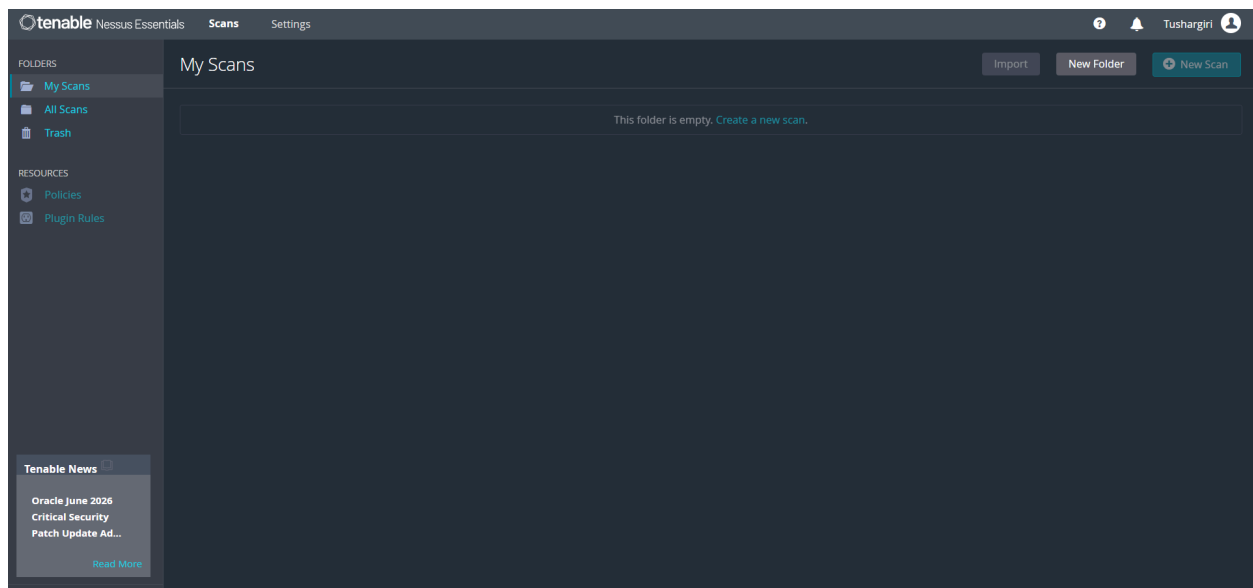
Installation Steps

1. Downloaded Nessus Essentials.
2. Installed the Nessus application.
3. Activated Nessus using the activation code.
4. Updated vulnerability plugins.
5. Accessed the Nessus web interface through:
<https://localhost:8834>
6. Completed initial setup and plugin synchronization.

Scan Policy Used

- Scan Type: Basic Network Scan
- Scanner: Local Scanner
- Severity Base: CVSS v3.0

Screenshot 1:



4. Scan Targets

Target 1 – Localhost

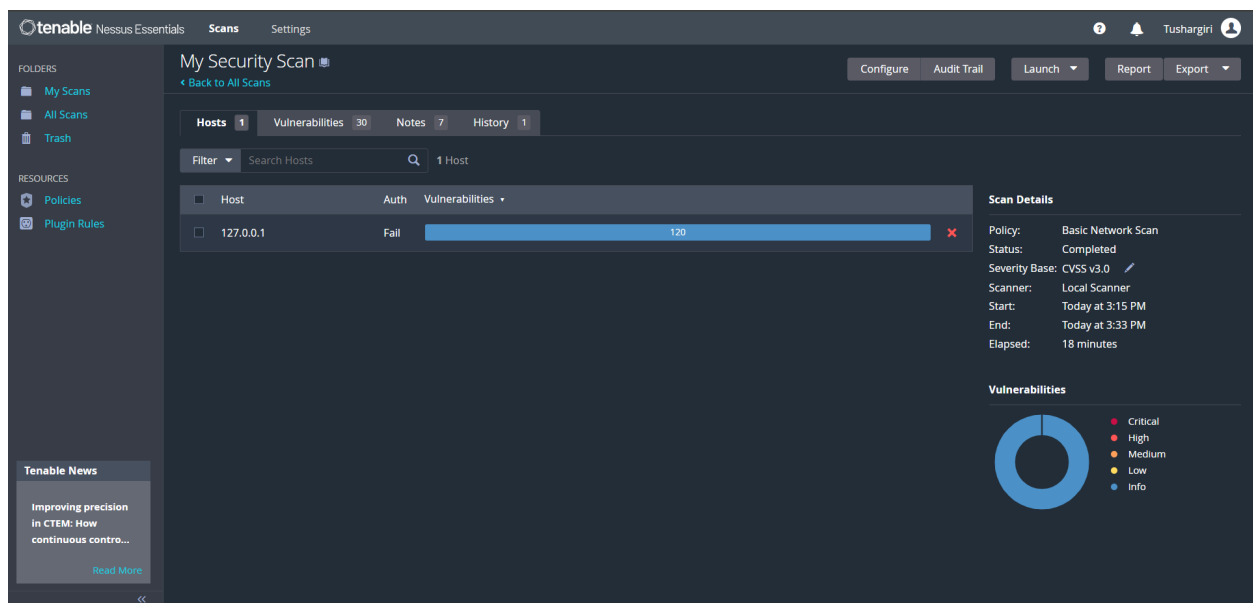
IP Address:

127.0.0.1

Purpose:

To assess the security posture of the local machine.

Screenshot 2:



Target 2 – Personal Test Machine(Windows)

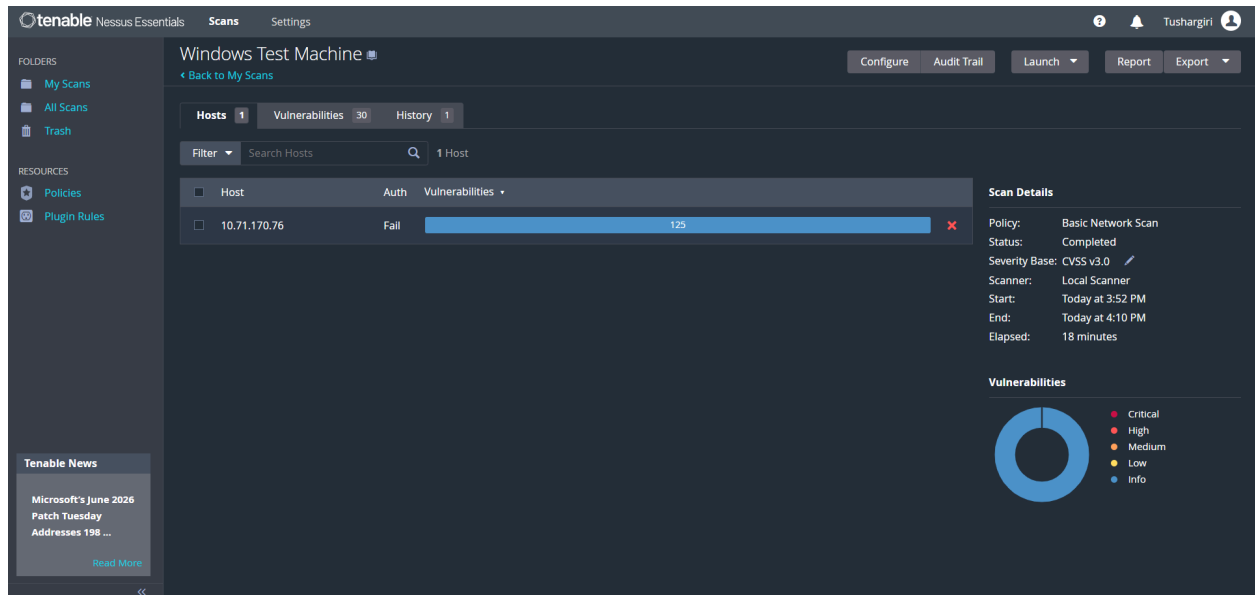
IP Address:

10.71.170.76

Purpose:

To identify services and vulnerabilities present on a Windows test machine.

Screenshot 3:

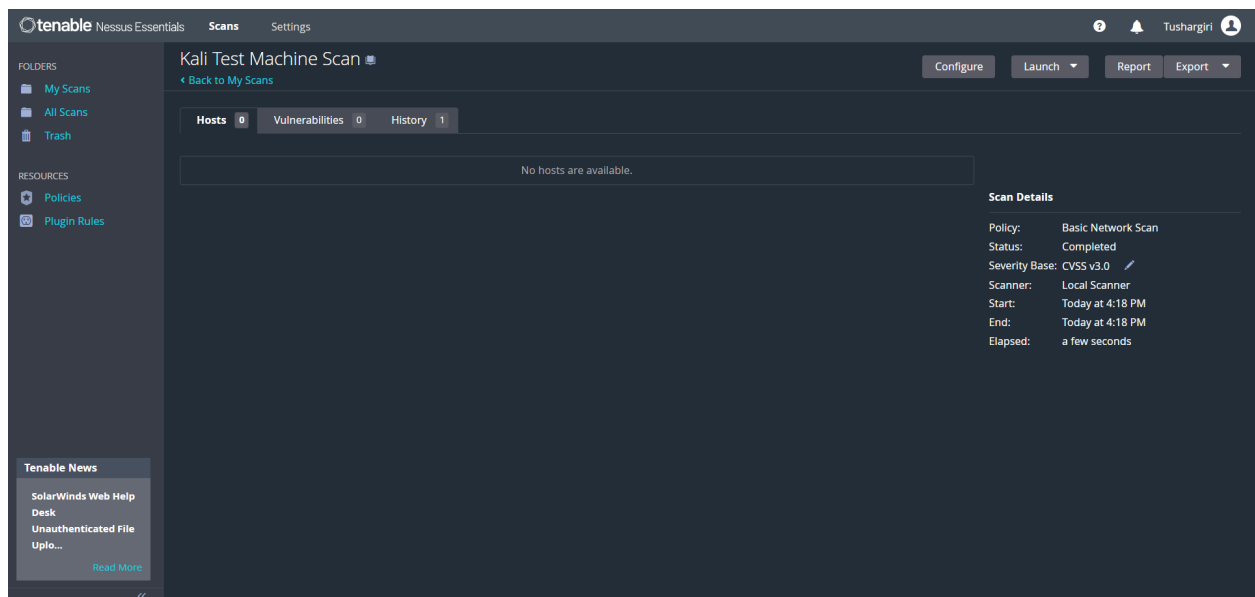


Target 3 – Lab Environment

Target:

Kali Linux Virtual Machine

Screenshot 4:



Observation:

The scan could not identify any reachable hosts because the Kali virtual machine was configured using VirtualBox NAT networking and was not reachable from the host system running Nessus.

5. Scan Results Summary

Severity Level	Findings
Critical	0
High	0
Medium	0
Low	0
Informational	30

Observation:

No Critical, High, Medium, or Low severity vulnerabilities were detected. The scan primarily identified informational findings related to active services, protocol configurations, and service detection.

6. Identified Services

The following services were detected during the assessment:

- HTTP Service
- SMB Service
- SSL Service
- TLS Service
- NetBIOS Service
- MySQL Service
- Windows Remote Services

These services were analyzed to determine their security configuration and potential risks.

7. Critical Findings

No Critical severity vulnerabilities were detected during the assessment.

8. High Findings

No High severity vulnerabilities were detected during the assessment.

9. Medium Findings

No Medium severity vulnerabilities were detected during the assessment.

10. Low Findings

No Low severity vulnerabilities were detected during the assessment.

11. Informational Findings

The following informational findings were detected:

HTTP Findings

- HyperText Transfer Protocol (HTTP)
- HTTP Server Type and Version
- HSTS Missing From HTTPS Server

Screenshot 5:

The screenshot displays the Tenable Nessus Essentials interface for a scan titled "Windows Test Machine / HTTP (Multiple Issues)". The interface includes a sidebar with folders (My Scans, All Scans, Trash) and resources (Policies, Plugin Rules). The main content area shows a table of vulnerabilities with columns for Severity, CVSS, VPR, EPSS, Name, Family, and Count. There are 3 vulnerabilities listed, all with an "INFO" severity level. A "Scan Details" panel on the right provides information about the scan policy, status, severity base, scanner, start/end times, and elapsed time. A "Vulnerabilities" donut chart at the bottom right shows the distribution of severity levels.

Sev	CVSS	VPR	EPSS	Name	Family	Count
INFO				HyperText Transfer Protocol (HT...	Web Servers	3
INFO				HTTP Server Type and Version	Web Servers	2
INFO				HSTS Missing From HTTPS Server	Web Servers	1

Scan Details

- Policy: Basic Network Scan
- Status: Completed
- Severity Base: CVSS v3.0
- Scanner: Local Scanner
- Start: Today at 3:52 PM
- End: Today at 4:10 PM
- Elapsed: 18 minutes

Vulnerabilities

- Critical
- High
- Medium
- Low
- Info

SMB Findings

- Microsoft Windows SMB Service Detection
- Microsoft Windows SMB Native LAN Manager Detection
- Microsoft Windows SMB Version Detection
- Microsoft Windows SMB2 and SMB3 Detection
- Windows NetBIOS / SMB Remote Host Information Disclosure

Screenshot 6:

The screenshot displays the Tenable Nessus Essentials interface for a scan titled "Windows Test Machine / SMB (Multiple Issues)". The interface includes a sidebar with folders (My Scans, All Scans, Trash) and resources (Policies, Plugin Rules). The main content area shows a table of vulnerabilities with columns for Severity, CVSS, VPR, EPSS, Name, Family, and Count. There are 5 vulnerabilities listed, all with an "INFO" severity level. A "Scan Details" panel on the right provides information about the scan policy, status, severity base, scanner, start/end times, and elapsed time. A "Vulnerabilities" donut chart at the bottom right shows the distribution of severity levels.

Sev	CVSS	VPR	EPSS	Name	Family	Count
INFO				Microsoft Windows SMB Service ...	Windows	2
INFO				Microsoft Windows SMB NativeL...	Windows	1
INFO				Microsoft Windows SMB Version...	Windows	1
INFO				Microsoft Windows SMB2 and S...	Windows	1
INFO				Windows NetBIOS / SMB Remot...	Windows	1

Scan Details

- Policy: Basic Network Scan
- Status: Completed
- Severity Base: CVSS v3.0
- Scanner: Local Scanner
- Start: Today at 3:52 PM
- End: Today at 4:10 PM
- Elapsed: 18 minutes

Vulnerabilities

- Critical
- High
- Medium
- Low
- Info

SSL Findings

- SSL Cipher Suites Supported
- SSL Perfect Forward Secrecy Cipher Suites
- SSL Cipher Block Chaining Cipher Suites

Screenshot 7:

The screenshot displays the Tenable Nessus Essentials interface for a scan titled "Windows Test Machine / SSL (Multiple Issues)". The left sidebar shows folders like "My Scans", "All Scans", and "Trash", along with resources like "Policies" and "Plugin Rules". The main panel shows a table of vulnerabilities with 30 total findings. The table columns include Severity, CVSS, VPR, EPSS, Name, Family, and Count. The right sidebar provides scan details and a vulnerability distribution donut chart.

Sev	CVSS	VPR	EPSS	Name	Family	Count
INFO				SSL Cipher Suites Supported	General	2
INFO				SSL Perfect Forward Secrecy Cip...	General	2
INFO				SSL Cipher Block Chaining Ciphe...	General	1

Scan Details

- Policy: Basic Network Scan
- Status: Completed
- Severity Base: CVSS v3.0
- Scanner: Local Scanner
- Start: Today at 3:52 PM
- End: Today at 4:10 PM
- Elapsed: 18 minutes

Vulnerabilities

- Critical
- High
- Medium
- Low
- Info

TLS Findings

- SSL/TLS Versions Supported
- TLS Supported Groups
- SSL/TLS Recommended Cipher Suites

Screenshot 8:

The screenshot displays the Tenable Nessus Essentials interface for a scan titled "Windows Test Machine / TLS (Multiple Issues)". The left sidebar shows folders like "My Scans", "All Scans", and "Trash", along with resources like "Policies" and "Plugin Rules". The main panel shows a table of vulnerabilities with 30 total findings. The table columns include Severity, CVSS, VPR, EPSS, Name, Family, and Count. The right sidebar provides scan details and a vulnerability distribution donut chart.

Sev	CVSS	VPR	EPSS	Name	Family	Count
INFO				SSL / TLS Versions Supported	General	2
INFO				TLS Supported Groups	General	2
INFO				SSL/TLS Recommended Cipher S...	General	1

Scan Details

- Policy: Basic Network Scan
- Status: Completed
- Severity Base: CVSS v3.0
- Scanner: Local Scanner
- Start: Today at 3:52 PM
- End: Today at 4:10 PM
- Elapsed: 18 minutes

Vulnerabilities

- Critical
- High
- Medium
- Low
- Info

12. Top 5 Vulnerabilities / Security Findings

1. HSTS Missing From HTTPS Server

Impact:

The web server does not enforce secure HTTPS communication.

Risk:

Attackers may attempt downgrade attacks or intercept unencrypted traffic.

Fix:

Enable HTTP Strict Transport Security (HSTS) to force browsers to use HTTPS.

2. HTTP Server Type and Version Disclosure

Impact:

Server software information is exposed.

Risk:

Attackers can identify server versions and search for known vulnerabilities.

Fix:

Hide server version banners and keep software updated.

3. SMB Version Detection

Impact:

The SMB version used by the system is disclosed.

Risk:

Older SMB versions may contain known security vulnerabilities.

Fix:

Disable SMBv1 and use SMBv2 or SMBv3.

4. SSL Cipher Suites Supported

Impact:

Supported encryption methods are revealed.

Risk:

Weak cipher suites can reduce communication security.

Fix:

Disable weak ciphers and allow only strong encryption algorithms.

5. SSL/TLS Versions Supported

Impact:

TLS protocol information is exposed.

Risk:

Older SSL/TLS versions may contain security weaknesses.

Fix:

Use TLS 1.2 or TLS 1.3 and disable deprecated protocols.

13. CVSS Information

CVSS (Common Vulnerability Scoring System) is used to measure the severity of vulnerabilities.

Observations:

- No vulnerabilities were assigned Critical severity.
- No vulnerabilities were assigned High severity.
- No vulnerabilities were assigned Medium severity.
- No vulnerabilities were assigned Low severity.
- Findings were informational and primarily related to service discovery and configuration information.

14. Remediation Recommendations

The following security improvements are recommended:

1. Enable HSTS on web servers.
2. Disable unnecessary service information disclosure.
3. Disable SMBv1 if enabled.
4. Use SMBv2 or SMBv3 only.
5. Enable strong SSL/TLS cipher suites.
6. Disable outdated SSL/TLS protocols.
7. Keep Windows and installed software updated.
8. Perform regular vulnerability assessments.
9. Restrict unnecessary network services.
10. Implement proper firewall rules.

15. Conclusion

Nessus Essentials was successfully installed and configured. Vulnerability scans were performed on localhost, a Windows test machine, and a lab environment. The assessment identified multiple informational findings related to HTTP, SMB, SSL, TLS, and system services. No Critical, High, Medium, or Low severity vulnerabilities were detected during the scans.

The results indicate that the scanned systems did not expose any known exploitable vulnerabilities under the selected scan policy. Regular vulnerability scanning, timely patch management, and secure service configurations should be maintained to ensure continued system security.